Flooding the filter: a game-theoretic model of AI, submission pressure, and quality collapse

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Abstract

Generative AI is transforming the nature of academic publishing. By September 2024, more than one-fifth of computer-science preprints showed evidence of large-language-model modification, while monthly submissions to arXiv reached a record 24,226, roughly 50% higher than in late 2022. Verification has not kept pace: reviewer acceptance rates have declined steadily over two decades, and between 6.5% and 16.9% of conference peer reviews are now themselves AI-modified. We study this asymmetry—the fact that AI scales paper production faster than it scales screening—using a game-theoretic model in which AI adoption raises submission volume and lowers drafting cost faster than it raises review capacity, while also increasing the rate at which low-quality manuscripts pass screening. We find three results. First, extreme slot scarcity can stabilize honest submission norms even under degraded screening, because the publication lottery becomes too hard to win. Second, gradual AI adoption can trigger a discrete tipping point into a low-quality long-run outcome; once tipped, the system is self-sustaining, so recovery requires a large coordinated intervention, not incremental reform. Third, expanding publication capacity without scaling verification accelerates the tipping. These results point to a concrete policy agenda of verification-centric institutions (structured audits, submission quotas, and decoupled certification) rather than simply more publication slots.

1 Introduction

We ask what happens to a publishing *system* when it becomes substantially cheaper and faster to produce a manuscript than to review one. By September 2024, more than one-fifth of computer-science preprints on arXiv showed measurable evidence of large-language-model modification (Liang et al., 2025), and at least 13.5% of all biomedical abstracts published that year contained AI-assisted text (Kobak et al., 2025). Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% higher than in late 2022 (arXiv, 2024). Verification has not kept pace. The rate at which researchers accept reviewer invitations has declined steadily over two decades (Meyerson et al., 2025), and between 6.5% and 16.9% of conference peer reviews now contain substantial AI-generated text (Liang et al., 2024). More than half of active researchers surveyed in 2025 reported using AI tools to write reviews, often against explicit journal policy (Naddaf, 2025). The question is not whether individual researchers behave ethically, but whether the system as a whole can absorb an asymmetric shock to production and verification.

While the conventional response to AI in science focuses on disclosure requirements, detection tools, and editorial guidelines, these measures address individual conduct and leave untouched a structural problem: even if every author is honest, reviewers can scrutinize only a finite number of manuscripts carefully per cycle. When submission volume rises faster than review capacity, screening degrades—not because of misconduct, but because of congestion. A reviewer who faces twice as many papers in the same time cannot give each submission the same attention. Congested screening lowers the effective quality threshold, and a lower threshold raises the expected payoff from submitting weak or low-quality work. We focus on this feedback between volume and screening quality as the central mechanism.

We distinguish two bottlenecks in the publishing system, and we argue that AI affects them asymmetrically. The first is verification capacity: the number of papers that reviewers can evaluate carefully in a given period, denoted K_{rev}. The second is allocation capacity: the number of papers the journal can publish per period, denoted K. AI raises submission volume V by reducing the marginal cost of drafting

and resubmitting papers, lowering the effort cost c of each submission. It may also increase the false-positive rate λ—the probability that a low-quality manuscript passes peer review—because AI-assisted writing produces polished text that is harder to distinguish from genuine contributions on surface features alone. Review capacity, by contrast, does not scale symmetrically. Deep expertise, data and code verification, and replication checking remain human-intensive tasks, and journal policies in many fields explicitly restrict AI use in reviews (Naddaf, 2025). Production, in short, scales faster with AI adoption than verification does.

We model this asymmetry in a game-theoretic framework, drawing on strategic models of scientific communication (Zollman et al., 2024; Thurner and Hanel, 2011). We consider a population of authors who choose between two strategies: submitting only genuine work (the cooperative strategy) or submitting everything including low-quality manuscripts (the defecting strategy). A journal applies peer review, subject to both a review capacity K_{rev} and a publication slot constraint K. We introduce an AI adoption intensity parameter $a \in [0, 1]$. Higher a raises submission volume V with elasticity γ, raises review capacity K_{rev} with elasticity $\delta < \gamma$, lowers submission cost c, and raises the false-positive rate λ. The production elasticity exceeds the verification elasticity by assumption, formalizing the asymmetry observed empirically. The net effect of a on the profitability of submitting a low-quality paper, $\Pi_{bad}(x, a) = r \cdot \bar{\rho}(x, a) \cdot \pi_0(x, a) - c(a)$, is positive: AI makes spamming more attractive.

We find three results. First, extreme slot scarcity can stabilize honest submission norms even under degraded screening, because the publication lottery becomes too hard to win. Scarcity, paradoxically, acts as a deterrent—but AI gradually erodes this protection by lowering the submission cost $c(a)$, so the deterrent requires progressively tighter slot constraints as adoption deepens. Second, gradual AI adoption can trigger a discrete tipping point into a low-quality long-run outcome; once tipped, the system is self-sustaining. Recovery requires a large coordinated intervention, not incremental reform: the value of a at which quality collapses is strictly lower than the value at which quality recovers, so the system exhibits hysteresis. Third, expanding publication capacity (raising K) without scaling verification makes the situation worse, not better. More slots raise the expected payoff from submitting marginal work; under AI adoption, where $c(a)$ is already low and $\lambda(a)$ already elevated, even a modest improvement in the publication lottery is enough to push the system across the tipping threshold.

These results point to a concrete policy agenda of verification-centric institutions—structured audits, submission quotas, and decoupled certification—rather than simply more publication slots. Effective intervention must target the two primary levers: raising genuine review capacity K_{rev} and reducing the false-positive rate λ. These are not incremental reforms; they constitute a different institutional design, one oriented toward verification rather than volume.

We proceed as follows. Section 2 documents the empirical asymmetry between AI-assisted production and verification. Section 3 reviews related literature. Section 4 presents the baseline model. Section 5 introduces the AI adoption parameter and derives the extended spam payoff. Section 6 analyses how AI adoption shifts equilibria, erodes the stability window, and amplifies the slot expansion paradox. Section 7 reports agent-based simulation results. Section 8 translates the analytical results into a four-principle institutional agenda and evaluates the mechanisms using Figure 10. Section 9 derives testable empirical predictions, proposes proxies for the key model channels, and outlines an identification strategy. Section 10 concludes.

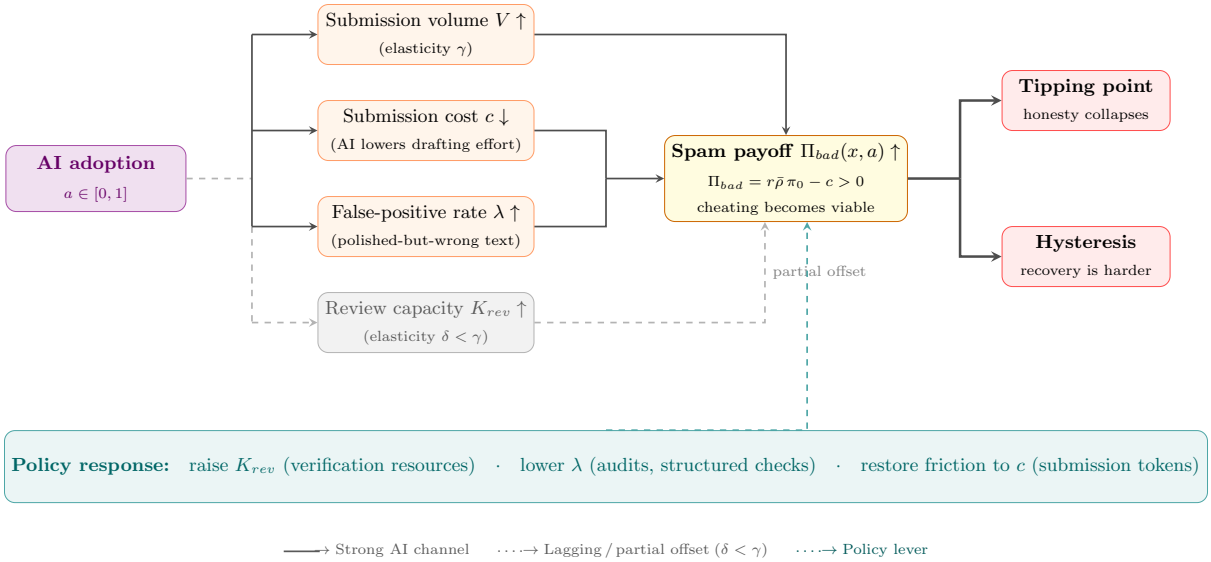

Figure 1: **Mechanism diagram: AI as an asymmetric shock to the publication system.** AI adoption (a) raises submission volume (V, elasticity γ) and lowers drafting cost (c), while polished AI text increases the false-positive rate (λ). Review capacity (K_{rev}) rises more slowly (elasticity $\delta < \gamma$), so the partial offset from AI-assisted reviewing is insufficient to prevent congestion. Together, these shifts raise the profitability of submitting low-quality work ($\Pi_{bad} = r\bar{\rho}\,\pi_0 - c$), pushing the system toward a tipping point and widening the gap between the collapse and recovery thresholds (hysteresis). Effective policy must raise K_{rev}, reduce λ, and restore friction to c (dashed teal arrow).

2 Stylized Facts: AI as an Asymmetric Shock

We draw on evidence that AI dramatically lowers the cost of producing a manuscript. Large language models have cut the time required to draft, revise, and resubmit academic text by a documented margin. In a randomised experiment with 453 college-educated professionals, Noy and Zhang (2023) found that access to ChatGPT reduced the time to complete an incentivised writing task by approximately 40% while simultaneously raising assessed output quality by 18 percentage points, with the largest gains accruing to weaker writers. Stokel-Walker and Van Noorden (2023) report that researchers can revise a full manuscript in under five minutes at an API cost below \$0.50. These micro-level reductions compound at the submission level: iterative resubmission across journals was once a significant labour cost, and AI converts that friction into a near-zero marginal activity, giving every author—regardless of the underlying quality of their work—a cost-effective reason to submit or resubmit. In our model, this channel corresponds to a downward shift in the submission cost parameter $c(a)$.

Aggregate data confirm a sharp rise in submission pressure. Macro-level data are consistent with the micro-level cost story. Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% above the late-2022 baseline; the first month to cross 20,000 submissions was May 2023, shortly after the public release of ChatGPT (arXiv, 2024). More than one-fifth of computer-science preprints sampled in September 2024 showed measurable evidence of large-language-model modification (Liang et al., 2025), and a vocabulary-based analysis of 15 million biomedical abstracts found that at least 13.5% of all 2024 *PubMed* papers contain AI-assisted text, with rates exceeding 40% in some sub-disciplines (Kobak et al., 2025). Alongside this volume growth, annual retraction counts exceeded 10,000 for the first time in 2023, with paper-mill activity implicated in mass retraction events at major publishers. These data show that submission volume V has shifted materially upward as AI adoption has spread, independently of any change in the fraction of authors who genuinely produce high-quality work. In our model, this corresponds to an upward shift in $V(x, a)$ with elasticity γ with respect to a.

While text production scales with AI, verification does not. Rigorous peer review requires reading carefully for logical consistency, checking data against claimed analyses, running or inspecting code, and evaluating genuine novelty—tasks that remain human-intensive in ways that text generation is not. Confidentiality requirements for submitted manuscripts prevent reviewers from uploading papers to third-party AI services, and many journals explicitly forbid AI use in the review process (Naddaf, 2025). Even where AI-assisted reviewing is nominally permitted, the marginal reviewer who can detect a subtle methodological flaw is a scarce specialist whose attention is not fungible with a language model's output. Reviewer acceptance rates have declined continuously for two decades (Meyerson et al., 2025), and the existing burden is already extreme: reviewers contribute more than 100 million hours annually to the peer review system worldwide, with the load concentrated among a small fraction of active researchers (Kovanis et al., 2016). We therefore expect that a system already under strain before the AI era cannot absorb a near-doubling of submission volume through a commensurate increase in review capacity. In our model, this channel is captured by $K_{rev}(a) = K_{rev}(0) \cdot (1 + \delta a)$, where the elasticity δ is strictly smaller than the production elasticity γ, reflecting the structural and policy constraints described above.

AI also polishes manuscripts without improving their reliability, which may raise the false-positive rate. A third channel operates through the quality of the screening signal rather than through volume. AI writing tools produce fluent, well-formatted, and stylistically consistent text (Stokel-Walker and Van Noorden, 2023), and this surface polish reduces the ability of reviewers to distinguish a low-quality submission from a genuine contribution on textual features alone, raising the probability that a flawed manuscript passes the review filter. The screening process degrades further when reviewers themselves turn to AI: between 6.5% and 16.9% of AI-conference peer reviews now contain substantial AI-generated text (Liang et al., 2024), and an experimental audit found that AI-assisted reviews award systematically higher scores, leading to elevated acceptance rates (Latona et al., 2024). Automated detection offers limited recourse—leading AI detectors exhibit false-negative rates above 70%, and adversarial paraphrasing can reduce recall to near zero (Liang et al., 2023). In our model, this channel corresponds to an upward shift in the false-positive rate $\lambda(a)$: as AI adoption rises, the probability that a low-quality submission passes review increases even holding the underlying composition of the author population constant.

We summarize these channels as a single asymmetry hypothesis. The four channels described above—rising submission volume, falling drafting cost, lagging review capacity, and rising false-positive risk—reduce to a single structural claim:

> *AI raises the elasticity of manuscript production above the elasticity of manuscript verification.*

We formalise this as the **AI asymmetry assumption**: with $a \in [0, 1]$ denoting AI adoption intensity,

$$\gamma > \delta \geq 0, \tag{1}$$

where γ is the elasticity of submission volume with respect to a and δ is the elasticity of review capacity with respect to a. This is not an axiom but a testable empirical statement about relative scaling rates. Table 1 translates each channel into a specific model parameter and states the direction of the shift. In Section 6, we examine how our qualitative results depend on the magnitude of the gap $\gamma - \delta$ and on the slope of $\lambda(a)$.

Table 1: **AI adoption channels and their model counterparts.** Each row maps an empirical mechanism to the corresponding model parameter, the direction of the shift, and the evidence base.

Empirical channel	Mechanism	Parameter	Shift	Key evidence
Lower drafting cost	AI reduces time and effort per manuscript; resubmission friction falls	$c(a)$	$\downarrow$	Noy and Zhang (2023); Stokel-Walker and Van Noorden (2023)
Higher submission volume	More authors submit more frequently; low-quality papers enter the pool	V (elasticity γ)	$\uparrow$	arXiv (2024); Liang et al. (2025); Kobak et al. (2025)
Lagging review capacity	Deep verification remains human-intensive; policy restricts AI in review; reviewer shortage pre-dates AI	K_{rev} (elasticity $\delta < \gamma$)	$\uparrow$ slowly	Meyerson et al. (2025); Kovanis et al. (2016); Naddaf (2025)
Rising false-positive rate	Polished AI text harder to distinguish; AI-assisted reviews inflate acceptance scores; detectors unreliable	$\lambda(a)$	$\uparrow$	Liang et al. (2024); Latona et al. (2024); Liang et al. (2023)
Possible quality uplift	AI may also help authors improve genuine contributions (partial offset)	α	$\uparrow$ (offset)	Noy and Zhang (2023)

3 Related Literature

We draw on research showing that peer review functions as a shared resource under increasing strain. Each submission consumes a fixed amount of reviewer time and attention, and when submission volume grows faster than the pool of willing reviewers, that resource degrades. Gross and Bergstrom (2019) show that scientific contests become inefficient at high participation rates: total effort rises, but expected quality per participant falls. Cotton (2013) reaches a parallel conclusion for journals: a small entry cost can deter low-quality submissions and relieve pressure on reviewers. Kovanis et al. (2016) estimate that the global peer review system already consumed over 100 million hours of researcher time each year in the mid-2010s, and Aczél et al. (2021) value the US contribution alone at more than 1.5 billion USD in 2020. Bianchi et al. (2018) model these pressures and find that screening quality declines even when reviewers act in good faith, simply because competing demands leave too little time per paper. The reviewing system, in short, was already near its limits before generative artificial intelligence entered the picture.

While congestion models treat reviewers as passive, a related body of work models authors as strategic agents—and this is where the feedback loop becomes visible. Thurner and Hanel (2011) show that when the review filter is imprecise, submitting everything regardless of quality can become individually rational, pushing the long-run outcome toward low average quality. Zollman et al. (2024) extend this to a multi-journal setting and find that prestige-maximising incentives can reward strategic over honest behaviour. Tiokhin et al. (2021) study honest signalling and show that the good outcome—where only strong work is submitted—is fragile: it collapses when the cost of mimicking quality falls even slightly. Smaldino and McElreath (2016) document this logic empirically, arguing that "publish or perish" incentives act as a selection force that spreads high-volume, low-quality strategies through academic communities. At the aggregate level, Ioannidis (2005) shows that most published findings fail to replicate, not because of fraud, but because structural incentives reward output over reproducibility. Casadevall and Fang (2014) and Stephan (2012) trace this to career and grant structures that treat paper counts as a proxy for scientific value. Paper mills—commercial services producing fabricated manuscripts to order—represent

the extreme end of this logic: annual retraction counts have reached record highs and single incidents have implicated thousands of papers at major publishers. Our model formalises the mechanism underlying these trends: when the review filter is weak and slots are available, submitting low-quality work yields a positive expected payoff, and selection pressure then favours volume over quality.

In contrast to the strategic submission literature, recent empirical work documents how generative artificial intelligence reshapes both sides of the system simultaneously. On the production side, Noy and Zhang (2023) find that access to a large language model cuts professional writing time by 40% and raises rated output quality by 18 percentage points. Liang et al. (2025) detect language-model modification in more than one-fifth of computer-science preprints by late 2024, and Kobak et al. (2025) find AI-assisted text in at least 13.5% of biomedical abstracts from the same period. On the evaluation side, however, the picture is more troubling. Checco et al. (2021) conclude that deep verification—checking data, code, and logical consistency—remains beyond current automated systems. Liang et al. (2024) find that between 6.5% and 16.9% of reviews at major conferences already contain AI-generated text, and Latona et al. (2024) show experimentally that AI-assisted reviews award higher scores, raising acceptance rates. Naddaf (2025) reports that more than half of surveyed researchers now use AI tools for reviewing, often against journal policy. These findings supply the empirical foundation for our asymmetry assumption: artificial intelligence raises submission productivity faster than it raises verification capacity.

A fourth strand addresses the economics of gatekeeping and institutional design. Ellison (2002) shows that the equilibrium publication standard depends on how strongly readers value quality signals. Muller-Itten (2022) finds that journals maintain strict standards to protect their certification value: accepting too freely devalues their own signal. Casadevall and Fang (2014) argue that effective reform must target the false-positive rate directly—through data deposition, structured reporting, and replication requirements—rather than adjusting slot counts alone. Ginsparg (2011) shows that dissemination and certification can be separated: preprint servers make manuscripts available immediately while editorial endorsement follows independently. Cotton (2013) provides the sharpest economic case for submission fees: a small cost of entry restores friction, filters marginal submissions, and reduces reviewer overload.

What we contribute

While these literatures each address part of the problem, none unifies them. Models of strategic submission do not treat artificial intelligence as a structural shock that shifts multiple inputs at once. The empirical AI-in-science literature documents faster production, slower review growth, and higher false-positive risk, but does not connect these to a model with clear predictions about the long-run state of the system. Neither strand addresses the possibility that submission pressure can grow gradually until it crosses a threshold, after which the system shifts suddenly to a low-quality state and resists recovery. The institutional design literature does not ask how its proposals must change when drafting cost approaches zero and the false-positive rate rises continuously with adoption. We close these gaps by connecting all four channels—$c(a)$ falling, V rising, K_{rev} lagging, and $\lambda(a)$ rising—to the long-run state of the publishing system, characterising conditions under which quality collapses suddenly and is hard to restore, and deriving which institutional levers can prevent or reverse that shift.

4 Baseline Model

Players and strategies. We consider a population of N authors submitting to a single journal over repeated periods. Each manuscript has binary quality: good ($q = 1$) with probability α, and bad ($q = 0$) with probability $1 - \alpha$. Standard dominance arguments reduce the strategy space to two options (Fudenberg and Tirole, 1991). Under **selective submission** (OG), an author submits only when $q = 1$. Under **universal submission** (AS), an author submits regardless of quality, hoping that imperfect screening allows bad papers through. We write $x \in [0, 1]$ for the share of authors playing OG; the remaining $1 - x$ play AS.

Two bottlenecks. The publication process has two distinct chokepoints; each creates a different strategic force. The first is **review capacity** K_{rev}: the number of papers that can receive careful scrutiny

per period. Submission volume is $V(x) = N[1 - (1 - \alpha)x]$, which rises as the AS share grows. When V exceeds K_{rev}, only a fraction of submissions are carefully reviewed; the rest pass without scrutiny—a rubber-stamping effect documented in overloaded systems (Azar, 2005; Kovanis et al., 2016). The **screening intensity** is:

$$\eta(x) = \min\left(1,\ \frac{K_{rev}}{V(x)}\right). \tag{2}$$

The second bottleneck is **publication capacity** K: the number of slots the journal can fill per period. Papers that pass review form a peer-accepted pool $M(x)$; at most K are published. The probability that a peer-accepted paper is ultimately published—the **rationing probability**—is:

$$\bar{\rho}(x) = \min\left(1,\ \frac{K}{M(x)}\right). \tag{3}$$

Screening errors and the spam payoff. Peer review is an imperfect filter (Bornmann et al., 2010; Cicchetti, 1991). A good paper is falsely rejected with probability ϵ; a bad paper wrongly passes with probability λ. Both error rates apply only to papers that are actually reviewed. When $\eta < 1$, unscreened papers pass automatically, raising the effective pass rates:

$$\pi_0(x) = 1 - \eta(x)(1 - \lambda), \qquad \pi_1(x) = 1 - \eta(x)\,\epsilon. \tag{4}$$

As $\eta \to 0$, both rates converge to 1: the journal loses all discriminatory power. The peer-accepted pool is $M(x) = N\alpha\,\pi_1(x) + N(1 - \alpha)(1 - x)\,\pi_0(x)$. Because OG and AS authors submit good papers on identical terms, the strategic difference reduces to the net value of submitting a bad paper:

$$\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c, \tag{5}$$

where r is the publication reward and c the submission cost. We work in the region $\lambda r < c < r(1 - \epsilon)$: without congestion, submitting a bad paper is unprofitable, but the incentive reactivates once screening degrades—the same adverse-selection logic as in Akerlof's market for lemons (Akerlof, 1970).

Dynamics. Strategy shares evolve according to the replicator equation (Weibull, 1995; Hofbauer and Sigmund, 1998):

$$\dot{x} = -x(1 - x)(1 - \alpha)\,\Pi_{bad}(x). \tag{6}$$

When $\Pi_{bad} > 0$, AS spreads and x falls; when $\Pi_{bad} < 0$, OG spreads and x rises. Two forces compete as the AS share grows. First, rising volume degrades screening (η falls, π_0 rises): submitting bad papers becomes more attractive. Second, rising volume floods the publication queue ($\bar{\rho}$ falls): the publication lottery becomes harder to win. When the screening-collapse force dominates, the system tips toward full corruption. When the rationing force dominates, extreme slot scarcity deters spam even after screening has failed—a mechanism we call *defence by congestion*.

Four long-run outcomes. The signs of Π_{bad} at the two boundary states determine which long-run outcome the system reaches. Table 2 summarises the four cases. A key structural observation is that the two boundaries are controlled by different parameters: the honest boundary ($x = 1$) is secured by keeping λ low, while the corrupt boundary ($x = 0$) is held by keeping K tight. This means the two bottlenecks can be targeted separately as policy instruments—a point we return to in Section 6. Figure 2 illustrates the bistable case. Π_{bad} is positive near $x = 0$ (corruption reinforces itself) and negative near $x = 1$ (honesty reinforces itself), with an unstable tipping threshold at $x^* \approx 0.82$ separating the two regions. Which long-run outcome prevails depends entirely on where the system starts relative to this threshold.

Table 2: **Four long-run outcomes determined by boundary payoffs.** $\Pi_{bad}(1)$ is the spam payoff when almost all authors are honest (low-volume limit); $\Pi_{bad}(0)$ is the spam payoff when almost no authors are honest (maximum-load limit).

Regime	$\mathbf{\Pi_{bad}(1)}$	$\mathbf{\Pi_{bad}(0)}$	**Long-run outcome**
GH	< 0	< 0	**Global honesty.** Spamming is unprofitable everywhere. The system reaches $x = 1$ from any starting point. Primary lever: keep $\lambda < c/r$.
GC	> 0	> 0	**Global corruption.** Spamming is profitable everywhere. The system collapses to $x = 0$ from any starting point. Seen in predatory journals with weak review and ample slots.
BS	< 0	> 0	**Bistability.** Both $x = 1$ and $x = 0$ are long-run outcomes. An unstable tipping threshold x^* separates the regions; history determines which outcome prevails. Recovery from corruption requires a large intervention.
SC	> 0	< 0	**Stable coexistence.** Neither extreme is reached. A stable interior outcome x^* emerges, sustained by rationing. Seen in top-tier journals where the publication lottery is very hard to win.

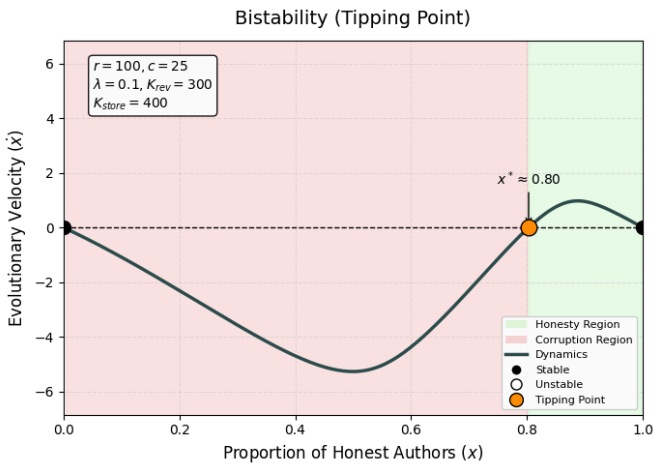

Figure 2: **Baseline phase portrait (bistability).** The curve plots $\Pi_{bad}(x)$, the net payoff from submitting a bad paper, against the honest-author share x. Parameters: $N = 1000$, $K_{rev} = 300$, $K = 400$, $r = 100$, $c = 25$, $\lambda = 0.1$. Red shading: $\Pi_{bad} > 0$, so AS spreads and x falls. Green shading: $\Pi_{bad} < 0$, so AS shrinks and x rises. Two long-run outcomes coexist—full honesty ($x = 1$) and full corruption ($x = 0$)—separated by an unstable tipping threshold at $x^* \approx 0.82$ (orange marker). Which outcome prevails depends entirely on the starting position relative to x^*.

5 The AI Extension

AI adoption as a parameter shift. Generative AI affects the publication system through multiple channels simultaneously. We capture this with a single scalar $a \in [0, 1]$, where $a = 0$ denotes the pre-AI baseline and $a = 1$ denotes full adoption across the author population. The parameter can be interpreted as the field-level intensity of AI use—a composite of the fraction of authors employing AI tools and the depth of their integration into the writing and submission workflow. We treat a as a comparative-statics parameter: we hold a fixed and trace how the equilibrium structure from Section 4 changes as a deepens. As established in Section 2 and illustrated in Figure 1, AI acts through four distinct channels, and the channels are not symmetric.

Channel 1—Production: more submissions at every composition. The most direct effect of AI is to lower the marginal cost of drafting and resubmitting a manuscript, which raises total submission volume at any given composition of the author population. Randomised experiments find that LLM access cuts professional writing time by roughly 40% and reduces resubmission effort to near zero (Noy and Zhang, 2023; Stokel-Walker and Van Noorden, 2023). We formalise this as a multiplicative shift in total submission volume:

$$V(x, a) \;=\; N\big[1 - (1 - \alpha)x\big] \cdot (1 + \gamma a), \tag{7}$$

where $\gamma > 0$ is the **production elasticity**. System-level data confirm this: monthly arXiv submissions rose roughly 50% in two years, with over one-fifth of preprints showing measurable LLM modification (arXiv, 2024; Liang et al., 2025). Crucially, V rises with a regardless of x: even a fully honest author population generates more submissions because each author drafts and resubmits more often.

Channel 2—Friction: lower cost of submitting bad papers. Beyond raising volume, AI directly lowers the cost of each submission attempt. We write:

$$c(a) \;=\; c_0(1 - \phi\, a), \qquad \phi \in (0, 1), \tag{8}$$

where c_0 is the baseline cost and ϕ governs the rate at which AI erodes submission friction. The cost reduction enters Π_{bad} directly: a lower c raises the net payoff from submitting a bad paper even if acceptance probabilities are unchanged. The effect is also asymmetric across strategies. For OG authors, who submit only when they have good work, cost reductions matter relatively less—they submit infrequently by choice. For AS authors, who submit everything, the lower c directly amplifies the incentive to flood the system.

Channel 3—Vulnerability: weaker screening signal. AI-generated text is harder to distinguish on surface features (Stokel-Walker and Van Noorden, 2023), AI-assisted reviews award systematically higher scores (Latona et al., 2024; Liang et al., 2024), and automated detection remains unreliable—false-negative rates exceed 70% (Liang et al., 2023). Each mechanism raises the probability that a bad paper passes review. We write:

$$\lambda(a) \;=\; \lambda_0 + \mu\, a, \qquad \mu \geq 0, \tag{9}$$

where λ_0 is the baseline false-positive rate and $\mu \geq 0$ is the sensitivity of screening quality to AI adoption. $\lambda(a)$ is a reduced-form parameter: it captures the net effect of AI adoption on the false-positive rate without modelling each mechanism separately. Rising λ enters $\pi_0 = 1 - \eta(1 - \lambda)$ directly: even a carefully reviewed paper is more likely to pass.

Channel 4—Review assistance: capacity grows, but more slowly. AI can also assist reviewers—summarising papers, flagging inconsistencies, checking statistical reporting—so review capacity is not fixed. We write:

$$K_{rev}(a) \;=\; K_{rev}(0) \cdot (1 + \delta\, a), \tag{10}$$

where $\delta \geq 0$ is the **verification elasticity**. However, deep verification remains human-intensive: confidentiality rules restrict AI use in reviews (Naddaf, 2025; Checco et al., 2021), and reviewer capacity was already strained before the AI era (Meyerson et al., 2025). These constraints enforce the asymmetry $\gamma > \delta \geq 0$ (Equation 1).

Figure 3 summarises the four channels and their functional forms.

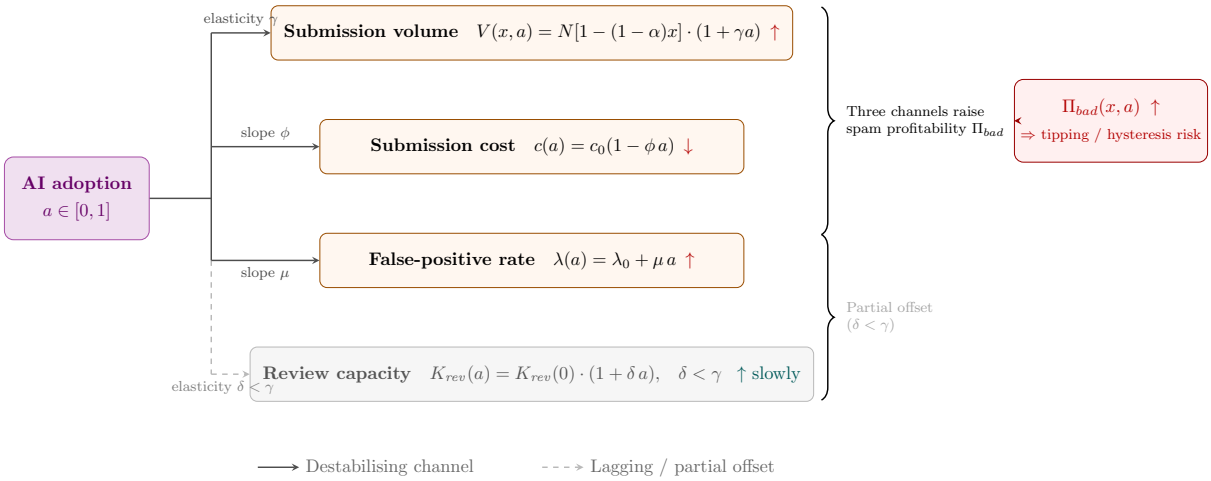

Figure 3: **AI adoption shifts four model parameters (Figure 3).** Solid arrows indicate channels that directly raise spam profitability Π_{bad}: rising submission volume (V, elasticity γ), falling submission cost (c, slope ϕ), and rising false-positive rate (λ, slope μ). The dashed arrow indicates the only offsetting channel—review capacity (K_{rev}, elasticity δ)—which grows more slowly than submission volume under the asymmetry assumption $\gamma > \delta$. All four channel equations are formal specifications of the informal channels described in Section 2.

The combined effect: spam becomes more profitable. Substituting the four channels into the baseline model yields the AI-extended spam payoff:

$$\Pi_{bad}(x,a) \;=\; r \cdot \bar{\rho}(x,a) \cdot \pi_0(x,a) \;-\; c(a), \tag{11}$$

where $\eta(x,a) = \min(1, K_{rev}(a)/V(x,a))$, $\pi_0(x,a) = 1 - \eta(x,a)(1-\lambda(a))$, and $\bar{\rho}(x,a) = \min(1, K/M(x,a))$. To isolate the mechanism most cleanly, consider the honest boundary $x = 1$, where only good papers are submitted and slot rationing is typically absent ($\bar{\rho}(1,a) \approx 1$ when publication slots exceed honest-author output):

$$\Pi_{bad}(1,a) \;\approx\; r \cdot \pi_0(1,a) \;-\; c(a). \tag{12}$$

Under the asymmetry $\gamma > \delta$, the screening intensity $\eta(1,a)$ falls as a rises. Combined with rising $\lambda(a)$ and falling $c(a)$, this increases $\pi_0(1,a) = 1 - \eta(1,a)(1-\lambda(a))$ while reducing submission cost. Provided $\gamma > \delta$, $\mu \geq 0$, and $\phi > 0$:

$$\frac{\partial \Pi_{bad}(1,a)}{\partial a} \;=\; r \underbrace{\frac{\partial \pi_0}{\partial a}}_{>0\ (\gamma>\delta)} \underbrace{-c'(a)}_{=c_0\phi>0} \;>\; 0. \tag{13}$$

The condition for Global Honesty—$\Pi_{bad}(1,a) < 0$—becomes harder to satisfy as a rises. When $\Pi_{bad}(1,a)$ crosses zero, the system exits the GH regime and enters bistability or Global Corruption.

Creation versus verification: why AI is structurally asymmetric. The four-channel decomposition shows that AI exerts pressure from both the supply and quality dimensions simultaneously. Channels 1 and 2 affect the supply side: they make it cheaper and faster to produce manuscripts. Channel 3 affects the screening side: it makes it harder to distinguish good submissions from bad ones at the review stage. Channel 4 is the only force working against the other three, but it is structurally constrained to grow more slowly. Writing is largely a syntactic task that language models perform well; evaluation is a semantic and domain-specific task requiring human expertise, data access, and contextual judgment (Checco et al., 2021). Institutions designed for the pre-AI era assumed rough parity between production and verification rates. That assumption no longer holds, and the asymmetry is not a temporary calibration issue—it reflects a fundamental difference in what AI can automate.

Special cases: when stability can be preserved. Not every AI adoption trajectory leads to quality collapse. Equation (12) shows that the Global Honesty condition can persist even at high a if two circumstances hold jointly. First, if $\delta \approx \gamma$—review capacity scales nearly as fast as submission volume—then $\eta(1, a)$ does not fall and the congestion channel is neutralised. This could arise if structured AI-assisted verification pipelines (automated code checks, data-consistency audits) allow reviewers to handle more submissions without sacrificing depth. Second, if $\mu \approx 0$—AI tools do not degrade screening reliability, perhaps because structured reporting requirements close the gap that polished AI writing opens—then $\lambda(a)$ does not rise. If both conditions hold simultaneously, the cost reduction alone is typically insufficient to cross the tipping threshold, because $c(a)$ falls gradually while the honesty condition $c(a) > r\lambda(a)$ can remain satisfied. Table 3 maps the four combinations of these two conditions onto qualitative predictions about regime direction. The critical implication is that the institutional response determines which cell the system occupies: a is largely exogenous, but δ and μ are policy choices.

Table 3: **Scenario matrix: qualitative regime outcomes as AI adoption a rises.** Two policy-sensitive parameters determine the trajectory. δ/γ: how fast review capacity keeps pace with submission volume (a measure of investment in verification infrastructure). μ: how fast AI adoption raises the false-positive rate (a measure of screening vulnerability). Each cell describes the dominant direction of regime change as a increases from zero.

	$\delta \approx \gamma$**: review keeps pace**	$\delta \ll \gamma$**: review lags severely**
$\mu > 0$**:** screening degrades	*Partial stability.* Review congestion is controlled, but rising λ still weakens the screening signal. GH region shrinks; bistability expands. Verification investment is necessary but not sufficient.	*Highest risk.* Both the congestion and false-positive channels are active. $\Pi_{bad}(1, a)$ rises rapidly. The system moves toward Global Corruption. Risk of irreversible tipping is high.
$\mu \approx 0$**:** screening holds	*Stable.* AI raises both production and verification at comparable rates; screening quality is maintained. GH region is preserved. Consistent with AI-augmented structured review under strong institutional controls.	*Conditionally stable.* Slot rationing may still provide defence by congestion if K is tight. Fragile: any expansion of publication slots or rise in r can tip the system toward bistability. Requires active management of K.

Testable predictions. The AI extension generates four empirically distinguishable predictions. First, quality degradation should be *nonlinear* in AI adoption: the system holds near the honest long-run outcome while a remains below the tipping threshold, then shifts abruptly rather than gradually. Smooth, continuous quality decline would be inconsistent with the bistable structure. Second, fields with higher AI adoption should show review-congestion signatures—longer turnaround times, rising desk-rejection rates, declining reviewer acceptance rates—as proxies for falling η (Meyerson et al., 2025). Third, acceptance rates should rise in fields with widespread AI-assisted reviewing, independently of any improvement in underlying research quality, as a signature of rising λ (Latona et al., 2024). Fourth, expanding publication slots should have *asymmetric* effects: before tipping, more slots increase access; after tipping, more slots accelerate quality collapse by raising the spam payoff Π_{bad}. We examine the regime consequences of each lever in Section 6.

6 Equilibria, Tipping, and Hysteresis under AI Adoption

6.1 Equilibrium Conditions and the AI Shift

The long-run outcome of the publishing system is determined by the signs of the spam payoff Π_{bad} at two boundary states. At the *honest boundary* ($x = 1$), almost all authors are selective; the question is whether a single deviating author can profitably flood the system. At the *corrupt boundary* ($x = 0$), almost all authors submit everything; the question is whether the system can self-correct. These two boundaries are governed by different parameters. As established in Section 4, the honest boundary is

secured by keeping λ low; the corrupt boundary is held by keeping K tight enough that the publication lottery deters spam even under degraded screening.

We define publication quality as the share of genuinely good papers among those accepted:

$$Q(x) \;=\; \frac{N\alpha\,\pi_1(x)}{M(x)} \;=\; \frac{\alpha\,\pi_1(x)}{\alpha\,\pi_1(x) + (1-\alpha)(1-x)\,\pi_0(x)}. \tag{14}$$

Since slot rationing applies uniformly to the peer-accepted pool $M(x)$, it does not alter the ratio of good to bad papers; quality is therefore determined entirely by the composition x at the long-run outcome. The four possible long-run outcomes—GH, GC, BS, SC—and their quality levels are summarised in Table 2.

Under AI adoption $a > 0$, four simultaneous shifts push $\Pi_{bad}(x, a)$ upward at every interior x. Rising submission volume (elasticity γ) and lagging review capacity (elasticity $\delta < \gamma$) lower the screening intensity $\eta(x, a)$, raising the bad-paper pass rate $\pi_0(x, a)$. Falling submission cost $c(a)$ directly raises the net payoff from each submission attempt. Rising false-positive rate $\lambda(a)$ further degrades the review filter even for papers that receive full scrutiny. The combined derivative (Equation 13) is strictly positive under the asymmetry $\gamma > \delta$: AI makes spamming more attractive at every composition of the author population. The analysis in this section traces the structural consequences of this shift for regime boundaries, stability windows, and recovery conditions.

We use two representative parameter sets throughout (Table 4). Set A ($K_{rev} = 300$, $K = 400$, $r = 100$, $c = 25$, $\lambda = 0.1$) represents a typical journal prone to bistability, where the review bottleneck is the binding constraint. Set B ($K_{rev} = 1200$, $K = 180$, $r = 150$, $c = 30$, $\lambda = 0.4$) represents a top-tier journal where extreme slot scarcity is the defining constraint. AI elasticity parameters are fixed at $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$ throughout.

Table 4: Parameter calibration for numerical analysis

Symbol	**Description**	**Set A (Baseline)**	**Set B (Top-tier)**
N	Author population	1000	1000
α	Good-paper probability	0.2	0.2
K_{rev}	Review capacity	**300**	**1200**
K	Slot capacity	**400**	**180**
r	Publication reward	100	**150**
c	Submission cost	25	**30**
λ	False-positive rate	0.1	**0.4**
ε	False-negative rate	0.05	0.01

Set A: review bottleneck dominates; primary failure mode is bistability. Set B: slot scarcity dominates; illustrates defence by congestion.

6.2 How AI Shifts Regime Boundaries

Figure 4 presents the core comparative-statics result. The left panel shows the baseline ($a = 0$) regime map across the (K_{rev}, K) parameter space. The right panel shows the same map under high AI adoption ($a = 0.8$). The comparison is unambiguous: the Global Honesty (green) region shrinks substantially, the Bistability (yellow) and Global Corruption (red) zones expand and shift left, and the Stable Coexistence (blue) region contracts. A journal safely in the GH zone at baseline may find itself in BS or GC at $a = 0.8$ without any change in its own institutional parameters.

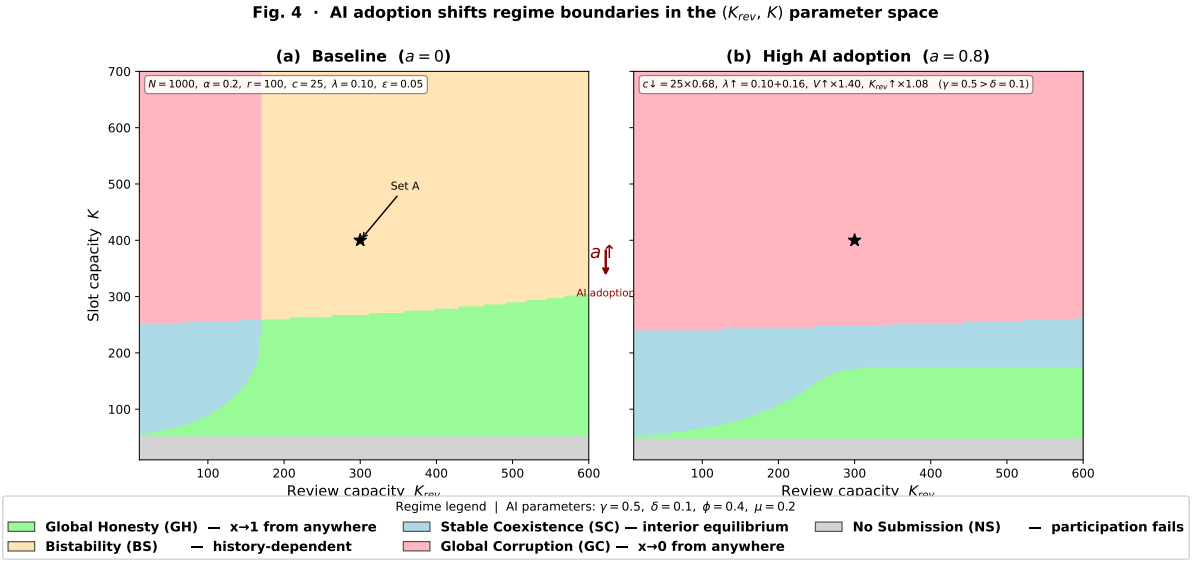

Figure 4: **AI adoption shifts regime boundaries in the (K_{rev}, K) parameter space (Fig. 4).** Left: baseline ($a = 0$). Right: high AI adoption ($a = 0.8$, $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$). The Global Honesty (green) region shrinks and the Bistability and Corruption zones expand. The Set A calibration point (star) moves from a safe location at baseline toward the boundary of GH and BS under high AI adoption. The asymmetry $\gamma > \delta$ is the key driver: review capacity grows too slowly to absorb the AI-driven surge in submission volume.

The magnitude of the shift is quantifiable. At $a = 0.8$ with $\gamma = 0.5$ and $\delta = 0.1$, submission volume has risen by 40% while review capacity has risen by only 8%. A journal with $K_{rev} = 300$ has the effective screening power of a journal with $K_{rev} \approx 231$ at baseline—a loss of nearly one-quarter of screening capacity with no change in institutional design. To restore the same screening intensity, the journal would need to raise K_{rev} to approximately 389, a 30% increase in review resources.

The slot expansion paradox is amplified by AI. Expanding publication slots (K) seems like an obvious response to submission pressure. Our model shows this response backfires when verification does not scale proportionally. Increasing K raises the rationing probability $\bar{\rho}(x, a)$, which raises the expected payoff from submitting a bad paper. This effect is amplified under AI adoption: when $c(a)$ is already low and $\lambda(a)$ is already elevated, even a modest improvement in the publication lottery is enough to push $\Pi_{bad}(1, a)$ from negative to positive, tipping the system from GH into Bistability. More slots increase the prize in a game whose entry barrier has already been eroded by AI. This is visible in Figure 4: moving upward (increasing K) in the right panel is more dangerous than the same move in the left panel. The slot expansion paradox identified in the baseline model is not merely preserved under AI adoption—it is exacerbated.

Defence by congestion and its limits. Extreme slot scarcity can stabilize an otherwise fragile system. When K is small relative to the peer-accepted pool M, the rationing probability $\bar{\rho}$ is so low that the expected payoff from submitting a bad paper falls below the submission cost even under degraded screening—the journal's publication lottery is simply too hard to win. This defence operates through the condition $\bar{\rho}(0, a) \cdot \pi_0(0, a) < c(a)/r$, and all three terms move against it as a rises. Rising $\pi_0(0, a)$ from degraded screening and rising λ increases the left side. Falling $c(a)$ reduces the right side. The net effect is that a journal relying on slot scarcity needs progressively tighter K to maintain the same deterrent effect as a increases. At some point, K must be so low that the participation constraint for honest authors—$\Pi_{good}(1, a) > 0$—is violated, pushing the system into the No Submission zone. Defence by congestion is a real stabilising force, but it is not robust to sustained increases in a.

6.3 Equilibrium Trajectories and Stability Window Erosion

Figure 5 traces the long-run honest-author share $x^*(a)$ as AI adoption rises from 0 to 1 under three parameter scenarios.

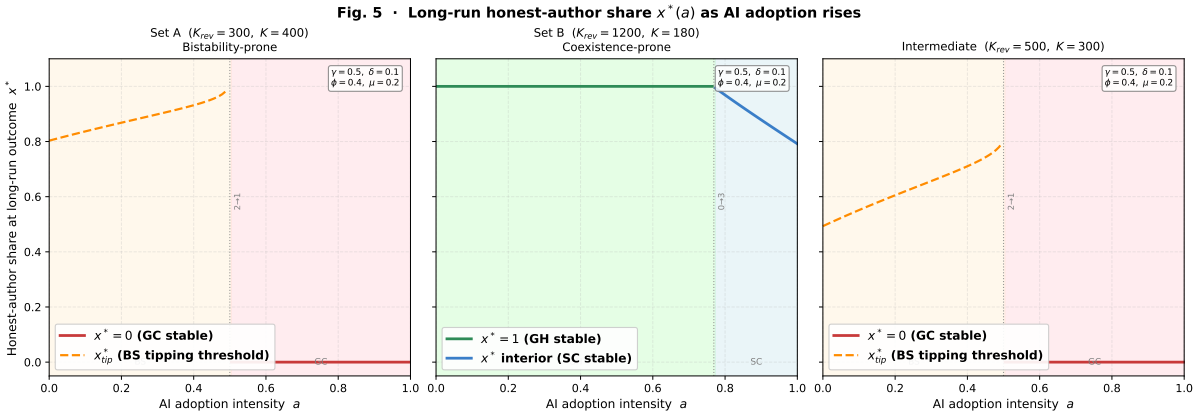

Figure 5: **Long-run honest-author share $x^*(a)$ as AI adoption rises (Fig. 5).** Left: Set A (bistability-prone). The system remains in GH ($x^* = 1$) until a crosses a threshold, then enters BS; the orange dashed curve shows the unstable tipping threshold x^*_{tip}, below which the system collapses. Centre: Set B (coexistence-prone). The interior stable long-run outcome x^* falls as a rises, tracking the declining profitability of honest submission relative to spamming. Right: Intermediate parameters. Shows the full GH→BS→GC transition as a increases. Vertical dashed lines mark regime transitions; shaded backgrounds indicate the active regime.

The left panel (Set A) shows a threshold effect. For low a, the system remains in GH: $x^* = 1$ and spamming is unprofitable everywhere. Once a crosses the critical value at which $\Pi_{bad}(1, a) = 0$, the system enters Bistability. The unstable tipping threshold x^*_{tip} (orange dashed) now separates two regions: above it, the honest norm is self-sustaining; below it, the system drifts toward full corruption. The practical implication is that the system can appear healthy—with x^*_{tip} far below the current share of honest authors—while becoming progressively more fragile. A single coordinated wave of low-quality submissions, or a field-wide shift in norms, can push the population below x^*_{tip} and initiate a collapse that the system cannot reverse on its own.

The right panel (Intermediate parameters) illustrates the full transition from GH through BS to GC as a increases. In the GC zone, the long-run outcome is $x^* = 0$: spamming dominates from every starting point. The path from GH to GC passes through a BS corridor, during which the long-run outcome is history-dependent and recovery is still possible. The width of this corridor—the range of a values during which recovery remains feasible—depends on how quickly $\lambda(a)$ rises and how much $c(a)$ falls.

Figure 6 maps this dynamic in the (N, a) plane, treating the author population size N as the external pressure variable.

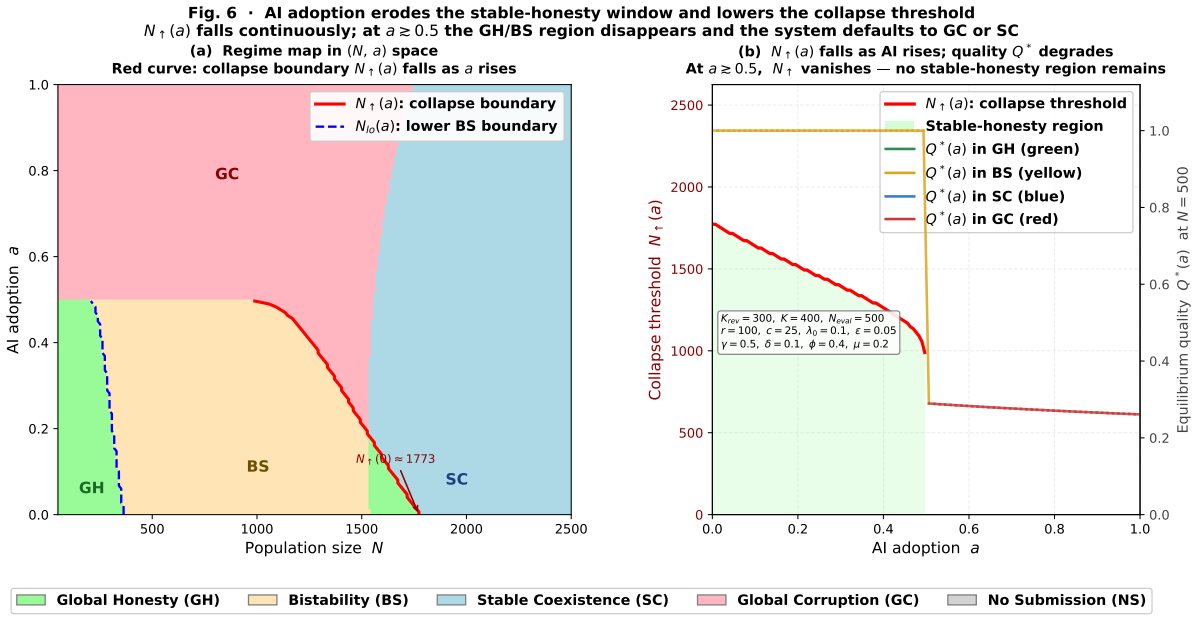

Figure 6: **AI adoption erodes the stable-honesty window and lowers the collapse threshold (Fig. 6).** Left: regime map in (N, a) space with Set A parameters. The red curve $N_\uparrow(a)$ marks the collapse boundary: to its right, the forward (honest-start) path collapses. The blue dashed curve marks the lower boundary of the Bistability zone. As a rises, $N_\uparrow(a)$ falls monotonically; at $a \gtrsim 0.5$ (for the chosen AI parameters) the GH and BS zones disappear entirely. Right: $N_\uparrow(a)$ (red, left axis) falls continuously; equilibrium quality $Q^*(a)$ at moderate pressure $N = 500$ (right axis, coloured by regime) also declines as a rises, tracking the regime transitions from GH (green) through BS (yellow) to SC (blue).

The left panel of Figure 6 shows the collapse threshold $N_\uparrow(a)$ falling monotonically as a rises. At $a = 0$, a population of $N \approx 1800$ is needed to push Set A out of the stable-honesty zone; at $a = 0.5$, this threshold has fallen to roughly half that. The right panel shows the corresponding quality decline: at a moderate load ($N = 500$), quality $Q^*(a)$ is near 1 at low a but falls continuously, tracking the regime transitions from GH through BS to SC. Crucially, at $a \gtrsim 0.5$, the bistable zone disappears entirely. There is no longer any starting position from which the honest norm is self-sustaining. Recovery from this state is not simply harder—it requires a qualitative change in institutional parameters (K_{rev} raised, λ reduced, c restored), not merely a reduction in pressure N. This is the sense in which AI adoption transforms the challenge from maintenance to repair.

6.4 The False-Positive Rate as Critical Destabiliser

Among the four AI channels, rising $\lambda(a)$ is structurally the most dangerous. The GH condition at $x = 1$ requires $\Pi_{bad}(1, a) < 0$; when screening is fully effective ($\eta = 1$), this simplifies to $\lambda < c(a)/r$. Under baseline parameters ($c = 25$, $r = 100$), this requires $\lambda < 0.25$. A slope of $\mu = 0.2$ over full adoption ($a = 1$) pushes λ from 0.10 to 0.30, crossing this threshold and making GH structurally impossible regardless of review capacity. No investment in K_{rev} can restore GH once $\lambda \geq c/r$, because even a perfectly reviewed bad paper passes at rate λ and yields a positive expected payoff. This result is consistent with signalling theory: the reliability of the screening signal—not merely the volume of screening—is what sustains the separating equilibrium (Tiokhin et al., 2021; Ellison, 2002).

The false-positive channel also explains why AI-assisted reviewing is particularly concerning. AI-assisted reviews have been shown to award systematically higher scores, raising acceptance rates (Latona et al., 2024; Liang et al., 2024). If widespread AI use in reviewing raises the average λ, the effect is direct and structural: the GH boundary shifts inward regardless of what authors do. Interventions that target λ directly—structured reporting checklists, mandatory data and code deposition, statistical audit requirements—therefore address the most critical vulnerability in the system.

6.5 Policy Levers and Summary Propositions

Each parameter corresponds to a concrete institutional intervention. We summarise the mapping in four propositions; formal derivations are in the Appendix.

Proposition 1 (Monotone destabilisation). *Under the asymmetry $\gamma > \delta \geq 0$, the spam payoff $\Pi_{bad}(x, a)$ is strictly increasing in a at every $x \in (0, 1)$. Consequently, the Global Honesty region in (K_{rev}, K) space shrinks weakly as a increases, and there exists a finite threshold $a^* \in (0, 1]$ above which no (K_{rev}, K) combination sustains GH without additional institutional intervention.*

Proposition 2 (Slot expansion paradox). *Holding K_{rev} and a fixed, increasing K weakly raises $\Pi_{bad}(x, a)$ at every x. The marginal destabilising effect of an additional slot is strictly larger at higher a, because lower $c(a)$ and higher $\lambda(a)$ amplify the gain from winning the publication lottery. The paradox is therefore amplified, not attenuated, by AI adoption.*

Proposition 3 (Erosion of congestion defence). *The corrupt-boundary stability condition $\Pi_{bad}(0, a) < 0$ requires $\bar{\rho}(0, a) < c(a)/\big(r \cdot \pi_0(0, a)\big)$. Since $c(a)$ falls and $\pi_0(0, a)$ rises with a, the required tightness of slot constraint increases with AI adoption. A journal that relied on passive scarcity for protection at baseline may need to actively reduce K to maintain the same deterrent as a rises.*

Proposition 4 (Policy lever ranking). *Among the primary levers, reducing λ (verification quality) and raising K_{rev} (review capacity) are unconditionally beneficial: they expand the GH region without side effects. Raising c (submission friction) restores the honesty condition but may deter some honest authors. Reducing K (tightening slots) provides passive deterrence but risks the No Submission zone. Expanding K alone is the only intervention that can make the system strictly worse.*

Early warning signals. The model identifies observable signatures that precede a tipping-point collapse. Rising submission-to-review ratios (V/K_{rev}) signal declining screening intensity η. Lengthening turnaround times and increasing desk-rejection rates indicate reviewer overload and the onset of rubber-stamping. Acceptance rates rising faster than improvements in underlying research quality signal a rising effective λ. In fields where AI adoption is high, these signals should appear before any observable quality decline, because the system can remain near $x \approx 1$ while the tipping threshold x^*_{tip} rises toward the current honest-author share. The gap between observable quality (still high) and structural fragility (rapidly growing) is precisely what makes AI-induced tipping difficult to detect in advance. We examine the testability of these predictions in Section 10.

7 Agent-Based Modeling

7.1 Why Simulate? Heterogeneity and the Diffusion of AI Adoption

The analytical results in Section 6 derive from a deterministic, infinite-population mean-field approximation (Hofbauer and Sigmund, 1998). Two features of real academic communities are suppressed by this approximation. First, AI adoption is not uniform: early adopters, late adopters, and non-adopters coexist within any field, and the overall adoption level a rises gradually through a diffusion process analogous to the spread of other general-purpose technologies (Rogers, 2003; Bass, 1969). Second, finite populations are subject to stochastic fluctuations that can push a system across a tipping threshold even when the deterministic drift would not (Nowak, 2006; Traulsen and Hauert, 2009). An agent-based model (ABM) allows us to test whether our three main results—gradual onset of fragility, discrete tipping, and severe path dependence—survive both of these complications (Bonabeau, 2002; Epstein and Axtell, 1996).

7.2 Model Design

We simulate a population of $N = 1000$ authors. Each author plays one of two strategies: selective submission (OG) or universal submission (AS); dominated strategies are excluded by the iterated elimination

result of Section 4. The journal's review capacity and quality parameters evolve with the field-level AI adoption level a_t, which rises exogenously along a linear ramp from $a = 0$ to $a = 0.8$ over the simulation horizon—a stylised representation of the diffusion trajectory documented empirically (arXiv, 2024; Liang et al., 2025). At each step, the effective parameters $K_{rev}(a_t)$, $\lambda(a_t)$, and $c(a_t)$ are updated according to the functional forms in Section 5. Strategy revision follows the pairwise Fermi comparison process (Szabó and Tőke, 1998; Traulsen and Hauert, 2009): a randomly chosen agent i compares its payoff Π_i with that of a randomly drawn peer j and switches with probability

$$P(s_i \to s_j) = \frac{1}{1 + \exp\big(-\beta\,[\Pi_j - \Pi_i]\big)}, \tag{15}$$

where $\beta = 1.0$ governs selection intensity. This rule captures bounded rationality: better-performing strategies spread, but noisily.

7.3 Results: Tipping as AI Adoption Rises

Figure 7 reports the core dynamic result. Panel (a) traces the tipping threshold $x^*(a_t)$—the unstable boundary separating the honesty and corruption basins—as a_t rises from 0 to 0.8. The threshold climbs from $x^* \approx 0.80$ at baseline to $x^* \approx 0.93$ under high adoption. The system appears healthy throughout: observed honest-author shares near the GH boundary remain at $x \approx 1$. Yet a fixed-size shock of $\Delta x = 0.18$—representing a coordinated wave of low-quality submissions, of the kind documented at major publishers during paper-mill incidents (Bik et al., 2016)—has opposite consequences depending on timing. Applied early at $a \approx 0.08$, it drops x to 0.82, which is still above $x^*(0.08) \approx 0.80$; the honesty norm self-repairs within a few hundred steps. Applied late at $a \approx 0.53$, the identical shock drops x to the same absolute level but now below $x^*(0.53) \approx 0.90$; the system collapses permanently to $x = 0$. The gap between x_t and $x^*(a_t)$—the *safety margin*—therefore shrinks silently even while observable quality remains high. This is precisely the "quiet before the cliff" signature that catastrophic-shift theory predicts for systems approaching a bifurcation (Scheffer, 2009).

Panel (b) confirms path dependence at $a = 0.45$ (bistable regime). Two trajectories separated by only $\Delta x_0 = 0.08$ diverge to opposite absorbing states: the run starting just above $x^*(0.45) \approx 0.92$ converges to full honesty within a few hundred steps, while the run starting just below collapses immediately. No endogenous force can reverse this collapse; the population is trapped in the low-quality long-run outcome (Schelling, 1978; Scheffer et al., 2001).

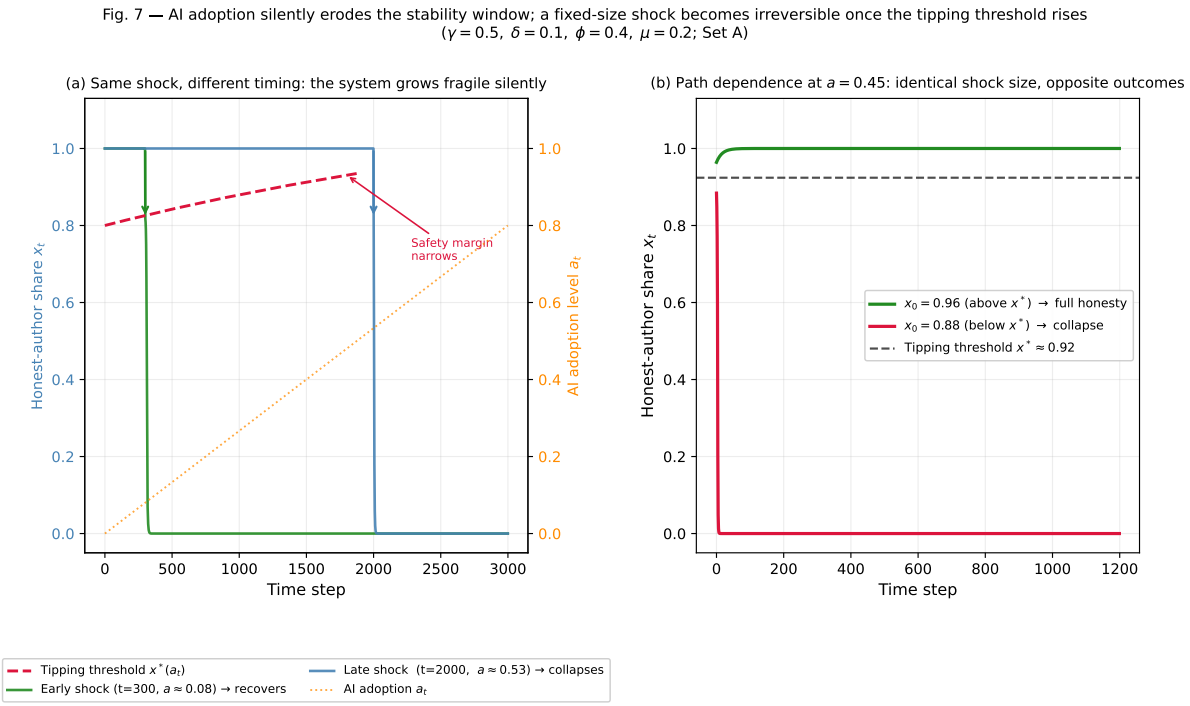

Figure 7: **AI adoption silently erodes the stability window (Fig. 7). (a)** The tipping threshold $x^*(a_t)$ (red dashed) rises from ≈ 0.80 to ≈ 0.93 as AI adoption increases. A fixed-size shock of $\Delta x = 0.18$ recovers when applied at $a \approx 0.08$ (green; safety margin intact) but causes permanent collapse when applied at $a \approx 0.53$ (blue; safety margin exhausted). Observable quality gives no warning: $x_t \approx 1$ in both cases until the shock hits. **(b)** Path dependence at $a = 0.45$: two initial conditions separated by $\Delta x_0 = 0.08$ diverge to opposite long-run outcomes. Parameters: Set A, $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$.

7.4 Robustness: Varying δ/γ and μ

Figure 8 maps the long-run outcome across the two policy-sensitive dimensions identified in Section 6: the verification elasticity ratio δ/γ and the false-positive slope μ. Both panels are evaluated at $a = 0.8$, representing sustained high AI adoption.

Panel (a) shows the long-run honest-author share x^* on a continuous colour scale. The parameter space is dominated by the collapse region ($x^* \approx 0$, dark red); a stable-honesty zone ($x^* \approx 1$, dark green) exists only in the bottom-right corner, where $\delta/\gamma \gtrsim 0.5$ and $\mu \lesssim 0.08$ simultaneously. Panel (b) confirms this pattern in terms of regime classification. The baseline calibration ($\delta/\gamma = 0.2$, $\mu = 0.2$, marked by white dotted lines) lies solidly inside the Global Corruption zone. The Bistability region (yellow) is confined to a narrow band at $\mu \lesssim 0.09$, independent of δ/γ. The Stable Coexistence and Global Honesty regions are accessible only when both conditions hold jointly; satisfying one alone is insufficient.

Two findings from this robustness check are particularly important. First, the μ dimension is the binding constraint: even if $\delta/\gamma = 1$ (review capacity scales perfectly with submission volume), the system remains in Global Corruption whenever $\mu > 0.09$. No investment in review staffing can compensate for a screening process that cannot distinguish polished AI text from genuine contributions (Checco et al., 2021; Liang et al., 2023). Second, the boundary between BS and GC is nearly vertical in the right panel, meaning that raising δ/γ from 0.2 to 0.5 shifts the system from GC to BS (where recovery is at least possible) only if μ is already low. This interaction confirms that the two policy levers—raising K_{rev} and reducing λ—must be deployed together, not sequentially.

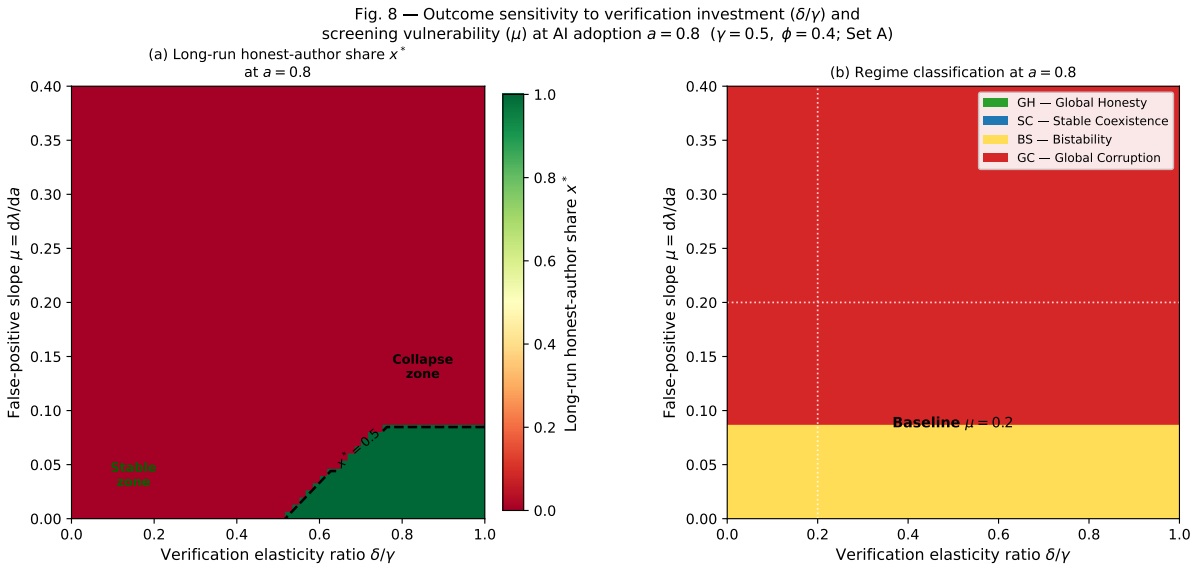

Figure 8: **Outcome sensitivity to verification investment and screening vulnerability (Fig. 8).** Both panels are evaluated at $a = 0.8$ across a grid of $\delta/\gamma \in [0, 1]$ and $\mu \in [0, 0.4]$. **(a)** Long-run honest-author share x^*: a stable-honesty zone (green) exists only when $\delta/\gamma \gtrsim 0.5$ and $\mu \lesssim 0.08$ simultaneously; the remainder of the space collapses ($x^* \approx 0$). **(b)** Regime classification: the baseline calibration (white dotted lines, $\delta/\gamma = 0.2$, $\mu = 0.2$) sits in Global Corruption. The Bistability corridor is narrow and confined to very low μ. Global Honesty requires high δ/γ and low μ jointly. Parameters: $\gamma = 0.5$, $\phi = 0.4$; Set A.

7.5 Stochastic ABM: Does Finite-Population Noise Validate the ODE?

The ODE assumes an infinitely large, well-mixed population and therefore suppresses all randomness. Real academic communities are finite. Even a system that sits nominally above the tipping threshold can cross it by chance when the number of authors is small (Traulsen and Hauert, 2009; Nowak, 2006). This subsection tests whether the deterministic tipping results of Section 6 survive under finite-population noise, and whether community size changes that verdict.

We extend the batch-matching ABM from Bonabeau (2002) to incorporate the same rising AI adoption schedule used in Section 7 (a_t linear from 0 to 0.8 over 3,000 steps). At each step t, the local screening capacity is $K_{rev,\text{eff}}(a_t) = K_{rev,0} \cdot (1 + \delta a_t)/(1 + \gamma a_t)$, the false-positive rate is $\lambda(a_t)$, and the submission cost is $c(a_t)$—identical functional forms to the ODE. The only departure from the ODE is that N is finite and strategy revision is stochastic: each revision attempt uses the Fermi rule (Szabó and Tőke, 1998), so the population does not move deterministically toward its expected next state but takes a noisy step. We run 30 independent replications for each of two community sizes: $N = 1{,}000$ (a full research field, the original calibration) and $N = 200$ (a specialised sub-field). Both start at $x_0 = 0.82$, which is 0.02

above the initial tipping threshold $x^*(a = 0) \approx 0.80$—a thin but nominally safe margin.

Figure 9 shows the results. Panel (a) overlays individual trajectories, the mean ABM path, and the ODE prediction. The ODE rises steadily to $x \approx 1$ and stays there: deterministic drift at $x = 0.82$ points toward honesty, and an infinite population never crosses the threshold by noise alone. The $N = 200$ runs tell a different story. Most collapse immediately; only a handful survive to the honest long-run outcome. The $N = 1{,}000$ mean settles near $x \approx 0.63$, reflecting the mix of collapsed and surviving runs in that sample. Panel (b) quantifies this: by $a \approx 0.06$, approximately 90% of $N = 200$ runs have already collapsed, while 35% of $N = 1{,}000$ runs have done so. Both fractions plateau for higher a, because the sorting between collapse and survival is complete before AI adoption has risen appreciably—the safety margin of only 0.02 is insufficient to resist even the earliest stochastic fluctuations.

Three conclusions follow from this comparison. First, the ABM confirms the ODE mechanism: in every run the tipping threshold $x^*(a_t)$ acts as the critical boundary, and no run escapes once it crosses below it. Second, the ODE systematically overestimates stability by ignoring finite-population noise. The same parameters that make the ODE predict "safe" lead to 35–90% stochastic collapse, depending on community size. This is consistent with theoretical results on stochastic evolutionary games: the probability of crossing a tipping threshold scales roughly as $\exp(-N \cdot \Delta)$, where Δ is the size of the safety margin, so thin margins are exponentially more dangerous in small populations (Traulsen et al., 2006). Third, community size is itself a policy variable. Fragmented peer-review ecosystems—specialised conferences, niche journals—face higher stochastic collapse risk than large generalist venues under the same AI adoption pressure. Institutional design should therefore calibrate the required safety margin $x_t - x^*(a_t)$ to the effective community size, not merely to the average field-level adoption level.

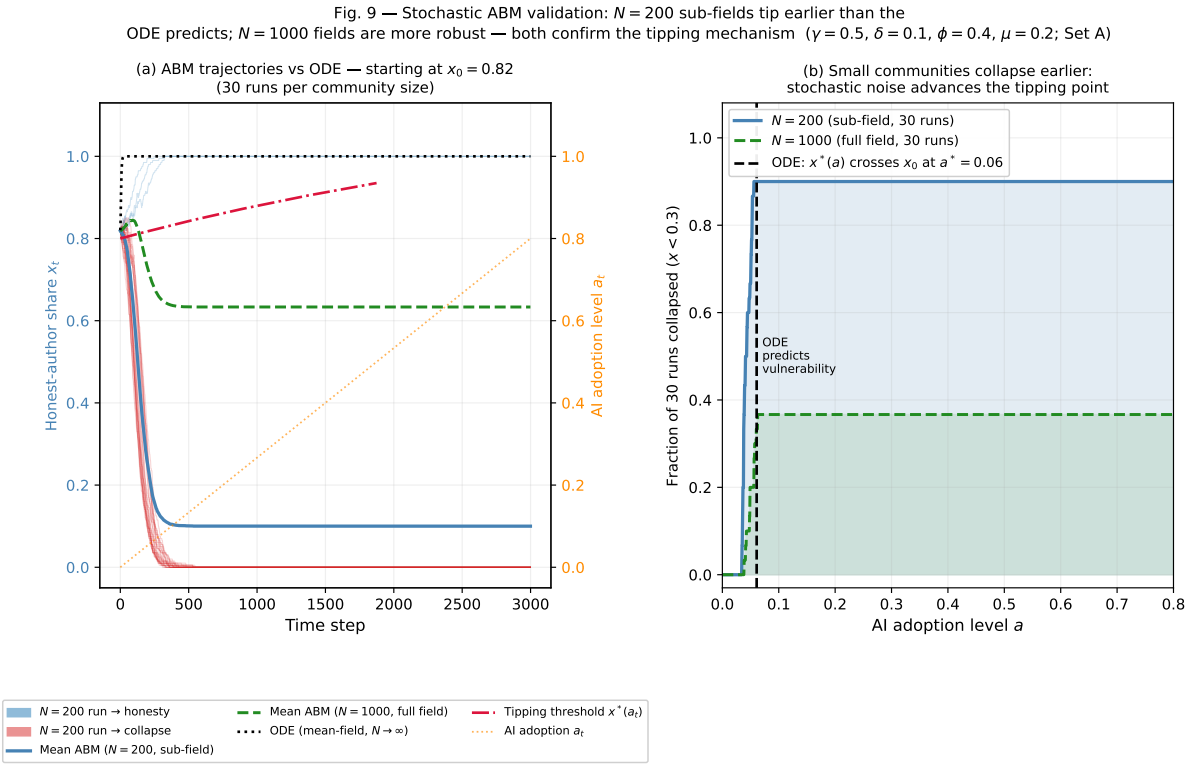

Figure 9: **Stochastic ABM vs ODE prediction (Fig. 9).** Both panels use Set A parameters with a linear AI adoption ramp, starting at $x_0 = 0.82$ (safety margin 0.02 above the initial tipping threshold). **(a)** Individual $N = 200$ run trajectories (blue: survive; red: collapse), mean paths for $N = 200$ and $N = 1{,}000$, the ODE prediction, and the rising tipping threshold $x^*(a_t)$. The ODE stays near $x = 1$; most $N = 200$ runs collapse within the first 200 steps. **(b)** Fraction of 30 runs collapsed ($x < 0.3$) as a function of a. By $a \approx 0.06$, 90% of $N = 200$ runs and 35% of $N = 1{,}000$ runs have collapsed—well before the ODE would predict vulnerability. Parameters: $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$; Set A.

8 Institutional Responses

The three results from Section 6 reduce the institutional problem to a single condition: keep $\Pi_{bad}(x, a)$ negative. AI adoption pushes Π_{bad} upward through four channels simultaneously: submission volume rises, review capacity lags, submission cost falls, and the false-positive rate λ increases. Effective policy must counteract each channel. We organize the response around four principles, ordered by the tightness of the constraint each relieves.

Principle 1: Scale verification ($K_{rev} \uparrow$). Higher review capacity raises the screening intensity $\eta = \min(1, K_{rev}/V)$, which reduces the bad-paper pass rate π_0 and lowers Π_{bad} at the corrupt boundary $x \approx 0$. Practical mechanisms include structured reviewer panels with explicit recognition credits (Aczél et al., 2021), AI-assisted consistency and format pre-screening that frees expert time for substantive judgment (Checco et al., 2021), and co-review programmes that train junior reviewers alongside senior ones. However, there is a binding caveat: when $\lambda > c(a)/r$, even perfect screening ($\eta = 1$) cannot make $\Pi_{bad}(1) < 0$ at the honest boundary. At $a = 0.8$ under baseline parameters, $\lambda = 0.26$ while $c(a)/r = 0.17$; no feasible increase in K_{rev} alone can stabilize the system at $K = 400$. Crucially, review assistance tools must not themselves raise λ: format and consistency checks are appropriate, but automated quality scoring is not (Naddaf, 2025).

Principle 2: Reduce the false-positive rate ($\lambda \downarrow$) — the binding constraint. The GH condition at the honest boundary requires $\Pi_{bad}(1) < 0$, which holds if and only if $\lambda < c(a)/r$. This is the binding constraint: no amount of K_{rev} expansion or slot restriction restores GH while λ exceeds this threshold. Three mechanisms directly reduce λ. First, mandatory data, code, and materials sharing makes fabrication costlier and enables post-publication verification (Munafò et al., 2017; Nosek et al., 2015). Second, structured reporting checklists—CONSORT for randomised trials (Moher et al., 2009), PRISMA for systematic reviews—force authors to document methods in ways that make gaps visible to reviewers. Third, Registered Reports commit authors to their hypotheses and analysis plan before results are known, eliminating outcome-selective reporting (Nosek et al., 2018), which is itself a leading source of inflated false-positive rates (Simmons et al., 2011). Taken together, these mechanisms address the root cause that λ is elevated: reviewers cannot easily distinguish genuine contributions from polished but uninformative manuscripts.

Principle 3: Add smart friction ($c \uparrow$). The GH condition also requires $c > r \cdot \lambda$. As AI adoption lowers $c(a) = c_0(1 - \phi a)$, this gap closes and eventually inverts. Submission tokens or annual per-journal quotas raise the effective cost of marginal submissions without requiring any detection of AI use (Cotton, 2013). Escalating resubmission fees impose additional costs on the low-quality cycling of rejected papers through multiple journals. We note an equity caveat: blunt friction raises costs for all authors, including those at under-resourced institutions; targeted exemptions are discussed in the final paragraph of this section.

Principle 4: Decouple dissemination from certification. The current publishing architecture conflates two distinct functions: disseminating findings quickly and certifying quality carefully. When certification degrades under submission pressure, both functions fail together. Separating the two—via preprint servers combined with overlay journals or community platforms such as Peer Community In (Ginsparg, 2011)—allows dissemination to remain immediate while certification proceeds at the pace required for genuine scrutiny. The slot-expansion paradox established in Section 6 applies specifically to certified publication slots K_{cert}, not to preprint availability. Decoupling therefore prevents the paradox from emerging: adding preprint capacity does not raise Π_{bad}, because preprint posting is not the payoff-generating event in the spam incentive.

Table 5 maps each mechanism to its primary parameter shift, and Figure 10 illustrates the joint effect of the principles in the (K_{rev}, K) parameter space. The left panel of Figure 10 shows that at $a = 0.8$ under baseline parameters, Global Corruption dominates most of the space, and Set A sits deep in the GC region; the navy arrow shows that raising K_{rev} alone (Principle 1) leaves Set A in GC because $\lambda = 0.26 > c/r = 0.17$. The right panel applies Principles 2 and 3 jointly ($\lambda = 0.05$, $c = 35$, with $\phi = \mu = 0$ locking in those improvements): the GC region is eliminated entirely and Set A shifts to the bistability (BS) region; the dark green arrow then shows that combining Principle 1 (K_{rev} raised to 600) carries Set A into the Global Honesty region. The lesson is that Principles 2 and 3 are necessary to exit GC, while Principle 1 converts bistability into global honesty: the three levers are complements,

not substitutes.

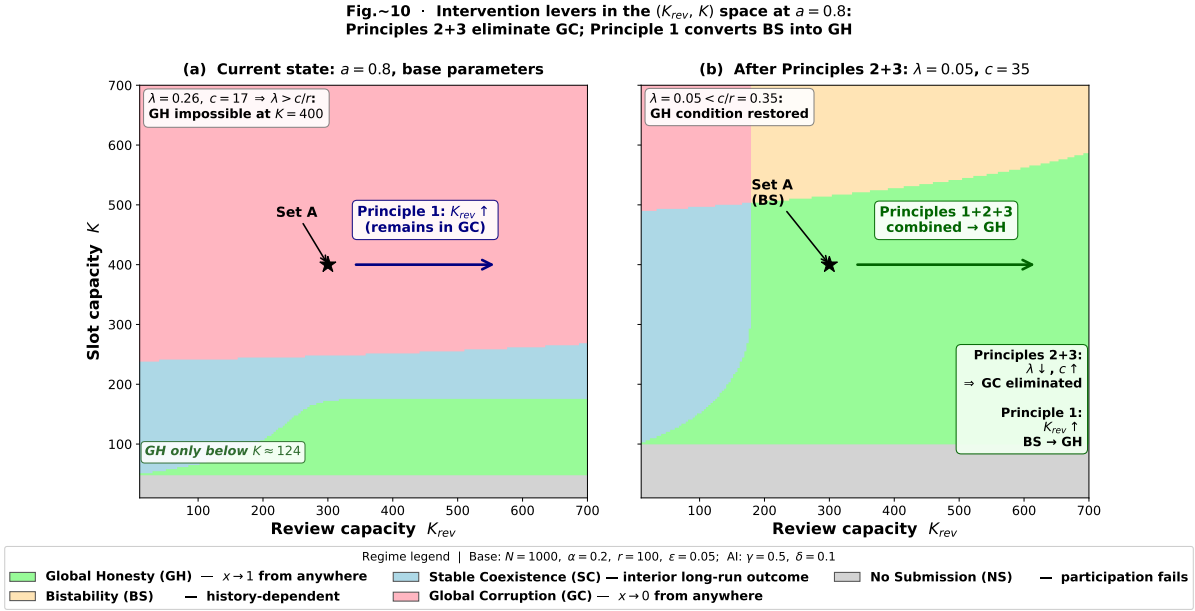

Figure 10: Intervention levers in the (K_{rev}, K) parameter space at $a = 0.8$. *Left panel:* base parameters ($\lambda = 0.26$, $c = 17$); GC dominates and Set A (star) sits deep in GC. The navy arrow shows Principle 1 ($K_{rev} \uparrow$) alone stays in GC because $\lambda > c/r$. *Right panel:* after Principles 2 and 3 ($\lambda = 0.05$, $c = 35$, $\phi = \mu = 0$); GC is eliminated and Set A shifts to BS. The dark green arrow shows the combined intervention (Principles 1+2+3, K_{rev} raised to 600) restores Global Honesty. Parameters: $\gamma = 0.5$, $\delta = 0.1$; Set A baseline $N = 1000$, $\alpha = 0.2$, $r = 100$, $\varepsilon = 0.05$.

Table 5: Institutional mechanisms and their model parameter effects. Each row maps a concrete institution to the parameter it primarily shifts. Arrows: ↑ raises, ↓ lowers, — no direct effect, ↑(·) partial or indirect effect. Publication reward r is unchanged by all mechanisms listed; adjusting r is a longer-run reform beyond the scope of this analysis.

Institution	Mechanism summary	K_{rev}	λ	c	K	Horizon
Principle 1: Scale verification						
Structured reviewer panels & recognition	Reward and recognise review effort	↑	—	—	—	Quick
AI-assisted pre-screening (format/consistency)	Flag obvious flaws before expert review	↑	—	—	—	Quick
Principle 2: Reduce false-positive rate						
Mandatory data, code, and materials sharing	Enables post-submission verification	↑(·)	↓	↑(·)	—	Quick
Structured reporting checklists (CONSORT/PRISMA)	Forces method transparency	—	↓	—	—	Quick
Registered Reports (pre-registered design)	Eliminates outcome-selective reporting	↑(·)	↓↓	↑	—	Medium
Dedicated statistical/methods editorial review	Specialist check of analytic choices	↑	↓	—	—	Quick
Principle 3: Add smart friction						
Annual submission tokens or quotas	Raise cost of marginal submissions	—	—	↑↑	—	Quick
Escalating resubmission fees	Target low-cost paper cycling	—	—	↑	—	Quick
Pre-submission triage/desk-rejection filter	Reject obvious mismatches before review	↑(·)	↓(·)	—	—	Quick
Principle 4: Decouple dissemination						
Preprint + overlay certification layer	Separate speed from quality guarantee	↑	—	—	↓(cert.)	Deep
Transition: post-collapse reset						
Coordinated field-wide mandate	Simultaneous cross-journal reform push	↑↑	↓↓	↑↑	—	Crisis

Transition policy: resetting after tipping. Hysteresis means that gradual, journal-by-journal reforms are insufficient once the system has tipped into Global Corruption. In the GC long-run outcome, $\Pi_{bad} > 0$ for all values of the honest-author share x; no individual author has an incentive to deviate back toward honesty (Scheffer et al., 2001). Recovery requires a large, coordinated intervention that simultaneously restores λ, raises c, and expands K_{rev} enough to shift the entire parameter configuration out of the GC region. The ClinicalTrials.gov registration mandate, which required prospective trial registration as a condition of journal publication, illustrates the scale of coordination needed (Moher et al., 2009): a single policy change, applied across many journals simultaneously, moved an entire field. The implication for AI-era publishing is that funders and professional societies must act across journal families, not journal by journal.

Equity. Friction mechanisms (c ↑) and strict verification requirements (λ ↓) impose larger burdens on authors at under-resourced institutions, who may lack the data-sharing infrastructure, statistical support, or time to meet demanding new standards. Smart design can preserve the incentive effects while redistributing the cost: submission token grants for authors at lower-income institutions (Cotton, 2013); structured reporting checklists, which carry low infrastructure cost, preferred over mandatory open data repositories, which carry high infrastructure cost; subsidized reviewer development programmes that build review capacity where it is currently thin (Aczél et al., 2021). These equity adjustments do not change the core model predictions—the regime boundaries shift on the same dimensions—but they change who bears the adjustment cost.

Policy menu: quick wins and deep redesign. Some mechanisms require no inter-journal coordination and can be adopted immediately: structured reporting checklists, mandatory data and code sharing policies, submission token caps, and escalating resubmission fees. Others require collective action across journals or fields: overlay certification architectures, cross-journal quota systems, and coordinated field-wide mandates. The right sequencing strategy is to deploy the quick wins first, to arrest the upward drift of Π_{bad}, while building the coordination capacity needed for deep redesign before any tipping event occurs. Waiting for a quality crisis before acting is the worst strategy: recovering from GC requires a larger simultaneous push than preventing the tip in the first place (Scheffer, 2009).

9 Empirical Predictions and Measurement

Prediction 1: Quality degradation is nonlinear—quiet, then a cliff. The bistable structure of the model generates a specific observable signature. Quality metrics should hold approximately steady while AI adoption a rises below the tipping threshold, then deteriorate rapidly once the tipping threshold $x^*(a)$ reaches the current honest-author share. Smooth, continuous quality decline would be inconsistent with the model: the bistable structure produces a cliff, not a slope. The early-warning indicators are not quality itself but the shrinking gap between the current author composition x_t and the rising tipping threshold $x^*(a_t)$—a gap that narrows silently while quality appears stable (Scheffer, 2009). Empirical evidence is consistent with this reading. Monthly arXiv submissions rose approximately 50% between late 2022 and late 2024, while reviewer acceptance rates and post-publication retraction counts have not yet stabilised at new long-run levels (arXiv, 2024; Van Noorden, 2023). Field-level heterogeneity provides additional structure: Kobak et al. (2025) find AI-modified text rates exceeding 40% in some biomedical sub-disciplines while remaining near zero in others, suggesting that different fields sit at different distances from their respective tipping thresholds—some already past it, others still in the quiet phase.

Prediction 2: The gap between collapse and recovery thresholds widens as AI adoption rises. As a increases, Proposition 3 shows that the defence-by-congestion mechanism requires progressively tighter slot constraints to hold the corrupt boundary, while Proposition 1 shows the Global Honesty boundary erodes. The model therefore predicts that the intervention required to restore quality in a field that has tipped is strictly larger than the intervention that would have prevented the tip. Pre-AI evidence establishes that the hysteresis mechanism itself is empirically real, independently of what triggers the initial collapse. Bik et al. (2016) document that image duplication—driven by paper mills, not generative AI—became entrenched in specific biomedical sub-fields despite sustained editorial intervention; Van Noorden (2023) report that annual retraction counts exceeded 10,000 for the first time in 2023 without receding, suggesting that the low-quality long-run outcome sustains itself once established. The pre-AI paper-mill literature thus confirms the mechanism: quality collapses are not self-correcting, and removing the proximate trigger (the mills) does not restore the parameter configuration that originally supported honest submission. The AI-specific prediction is that the same mechanism will be triggered at lower adoption thresholds and across more fields simultaneously, because AI lowers $c(a)$ and raises $\lambda(a)$ for the entire author population rather than for a subset of bad actors. A direct test would compare the institutional effort required to restore quality indicators in a field that tipped—whether by paper mills before 2022 or by AI-driven flooding after—against one that never tipped but faces equivalent submission pressure.

Prediction 3: Proxies for a rising false-positive rate. A rising λ produces a detectable mismatch between surface quality and substantive reliability: manuscripts become more polished while methodological soundness does not improve proportionately. Three observable proxies track this channel. First, **acceptance rates** should rise faster than any improvement in underlying research quality, because easier-to-pass manuscripts inflate the acceptance pool without proportionate quality gains; Latona et al. (2024) document this experimentally—AI-assisted reviews award systematically higher scores. Second, **surface linguistic quality** should improve independently of replication success or citation impact. Kobak et al. (2025) show that AI modification is measurable through excess vocabulary, providing a continuous field-level proxy for surface polish that can be compared against downstream quality indicators. Third, **replication rates** should decline in fields with high AI adoption if the screening signal

is genuinely degrading. The Open Science Collaboration found that fewer than half of 100 psychology studies replicated successfully (Open Science Collaboration, 2015); replicating that audit with papers published after 2022 in high-AI-adoption fields, and comparing against low-adoption fields, would directly test whether the false-positive channel is active.

Prediction 4: Proxies for review-capacity stress. The model predicts declining screening intensity $\eta = \min(1, K_{rev}/V)$ as submission volume grows faster than review capacity. Four observable proxies track this channel. First, **reviewer invitation acceptance rates** should decline faster in high-AI-adoption fields; Meyerson et al. (2025) document a continuous 21-year decline at a single journal, and the model predicts this trajectory steepens as AI-driven submission pressure intensifies. Second, **editorial turnaround times** should lengthen as editors wait longer for reviewer responses, signalling scarcity in the review pool. Third, **desk rejection rates** should rise as editorial offices pre-filter to protect reviewers from overload; an increasing share of submissions receiving no substantive evaluation is a direct proxy for η falling toward zero for that fraction of papers. Fourth, **reviewer concentration** should increase—a growing Gini coefficient on review counts across the active reviewer pool indicates that a shrinking fraction of researchers is absorbing a rising fraction of the total burden, a pattern that preceded the current shortage (Kovanis et al., 2016).

Prediction 5: Data sources and identification strategy. Testing these predictions requires field-level panel data on both the production and verification sides of the system. Several existing sources can be combined. On the production side, arXiv submission records are public, monthly, and tagged by subfield; the LLM modification classifier of Liang et al. (2025), applied to these records, provides a continuous field-level estimate of a_t. The vocabulary-based detection approach of Kobak et al. (2025), applied across the 15-million-abstract PubMed archive, offers an independent measure of AI modification at the sub-discipline level. On the verification side, reviewer acceptance rate series can be constructed from editorial logs at journals that publish annual statistics; acceptance rates, turnaround times, and retraction counts are available at scale through OpenAlex, CrossRef, and the Retraction Watch database. The natural experiment for causal identification is the release of ChatGPT in November 2022, which created a sharp and field-heterogeneous shock to submission cost c. A difference-in-differences design would compare fields with high ex-ante AI amenability—measured by pre-2022 adoption of computational tools, large collaborative networks, or text-heavy output—against low-amenability fields, before and after November 2022. The prediction is that high-amenability fields show faster growth in submission-to-review ratios, rising acceptance rates, and more rapid accumulation of post-publication retractions relative to low-amenability fields, controlling for field size and pre-shock submission trends.

Prediction 6: What would falsify the asymmetry hypothesis. The model's conclusions rest on two empirical claims: production scales faster than verification ($\gamma > \delta$), and the false-positive rate rises with AI adoption ($\mu > 0$). Either claim failing empirically would revise the institutional implications. If $\delta \approx \gamma$—if AI-assisted reviewing tools raise review capacity at nearly the same rate as AI raises submission volume—the congestion channel would not activate. This would be observable as stable submission-to-acceptance ratios and constant turnaround times despite rising submission volumes in high-adoption fields. If $\mu \approx 0$—if AI tools improve screening reliability, perhaps because structured AI-assisted checklists close the gap that polished writing opens—the false-positive channel would be neutralised. This would be observable as stable or falling acceptance rates in high-AI-adoption fields combined with stable replication rates for newly published work. The robustness analysis in Figure 8 shows that μ is the binding parameter: even if δ/γ rises toward 1, the system remains in Global Corruption whenever $\mu > 0.09$. A finding that AI-assisted reviewing consistently *improves* score calibration—rather than inflating it—would therefore be the strongest possible disconfirmation of the model's policy conclusions (Checco et al., 2021). The available evidence currently points in the opposite direction (Latona et al., 2024; Liang et al., 2024; Naddaf, 2025).

10 Conclusion

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